

Internal energy and The First Law

Internal energy: Energy not kinetic or potential

- 1) thermal energy (heat)
- 2) mechanical/strain energy
- 3) Chemical energy
- 4) Magnetization

Internal energy U of Ω_t : $U[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t) u(\mathbf{x}, t) dV_{\mathbf{x}}$

$u(\mathbf{x}, t)$ = internal energy density / unit mass

First Law of Thermodynamics

$$\boxed{\frac{d}{dt} U[\Omega_t] = Q[\Omega_t] + W[\Omega_t]} \quad \text{for all } \Omega_t \in \mathcal{B}_t$$

Net heating: $Q[\Omega_t] = \text{rate of thermal energy gain/loss}$

Net working: $W[\Omega_t] = \text{rate of mechanical energy gain/loss}$

$\Rightarrow$ heating first

Net heating

2 forms of heat gain/loss:

I) Body heating: $Q_b[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t) r(\mathbf{x}, t) dV_{\mathbf{x}}$

$r(\mathbf{x}, t)$ = heat supply/loss per unit mass

Example: radiogenic heating

$$r = H_0 e^{-\lambda t}$$

H_0 = initial power per unit mass

λ = decay constant

II) Surface heating: $Q_s[\Omega_t] = - \int_{\partial\Omega_t} \mathbf{q}(\mathbf{x}, t) \cdot \underline{\mathbf{n}}(\mathbf{x}, t) dA_{\mathbf{x}}$

$\mathbf{q}(\mathbf{x}, t)$ = heat flux vec or.

Fourier's law: $\mathbf{q} = - \underline{\underline{\kappa}} \nabla T$ (Lecture 10)

$\underline{\underline{\kappa}}$ = thermal conductivity tensor

Net heating is the rate at which heat is added or lost from Ω_t

$$Q[\Omega_t] = Q_b[\Omega_t] + Q_s[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t) r(\mathbf{x}, t) dV_{\mathbf{x}} - \int_{\partial\Omega_t} \mathbf{q}(\mathbf{x}, t) \cdot \underline{\mathbf{n}}(\mathbf{x}, t) dA_{\mathbf{x}}$$

Energy balance with net heating

First law with $W=0$: $\frac{d}{dt}U[\Omega_t] = Q[\Omega_t]$

where $U[\Omega_t] = \int_{\Omega_t} \rho u \, dV_x$

$$Q[\Omega_t] = \int_{\Omega_t} \rho r \, dV_x - \int_{\partial\Omega_t} \underline{q} \cdot \underline{n} \, dA_x$$

substituting:

$$\frac{d}{dt} \int_{\Omega_t} \rho u \, dV_x = - \int_{\partial\Omega_t} \underline{q} \cdot \underline{n} \, dA_x + \int_{\Omega_t} \rho r \, dV_x$$

using derivative relative to mass and divergence Thm

$$\int_{\Omega} (\rho \dot{u} + \nabla_x \cdot \underline{q} + \rho r) \, dV_x$$

by the arbitrariness of Ω_t we have

$$\boxed{\rho \dot{u} + \nabla \cdot \underline{q} = \rho r} \quad \text{local Eulerian form}$$

Following lin. mom. balance $\rightarrow$ conservative form

$$\boxed{\frac{\partial}{\partial t}(\rho u) + \nabla \cdot [\underline{v} \rho u + \underline{q}]} = \rho r \quad \Rightarrow \text{HW 9}$$

- still in terms of internal energy density!

Relate u to measurable quantities

Fundamental thermodynamic relation: $U = U(S, V)$

$$dU = T dS - p dV \quad \text{thermo book version}$$

U = internal energy [J]

S = entropy [$\frac{J}{K}$] V = volume [m^3]

T = temperature [K] p = pressure [Pa]

$\Rightarrow$ good for global statements (whole system)

Continuum mechanics is local

$u = \frac{U}{m}$ = specific internal energy [$\frac{J}{kg}$]
energy/mass m = mass

need $u = u(T, p)$

$$du = c_p dT - \frac{\alpha T}{\rho} dp \quad \rightarrow \text{take a good thermo class.}$$

c_p = specific heat capacity [$\frac{J}{kg K}$]

ρ = mass density [$\frac{kg}{m^3}$]

$\alpha = -\frac{1}{\rho} \left. \frac{\partial \rho}{\partial T} \right|_p$ = isobaric thermal expansivity [$\frac{1}{K}$]

Advection - Conduction Equation

Energy balance: $\frac{\partial}{\partial t}(\rho u) + \nabla \cdot [\underline{v} \rho u + \underline{q}] = \rho r$

$\underline{v} \rightarrow$ momentum balance

ρ, r parameters (given)

$u, \underline{q}$ unknown variables

1 equation with 4 unknowns

$\Rightarrow$ need to relate u and $\underline{q}$

Assume: $p = \text{const.}$ and $c_p = \text{const.}$ (HW9 $\rightarrow$ variable p)

Thermodynamics: $du = c_p dT$ ($dp = 0$)

$$u = u_0 + c_p (T - T_0) \quad c_p = \text{const.}$$

Fouriers law: $\underline{q} = -\kappa \nabla T$ (Lecture 10)

physical parameters:

1) $c_p = \text{specific heat capacity}$ $\left[\frac{\text{J}}{\text{kg K}} \right]$

2) $\kappa = \text{thermal conductivity}$ $\left[\frac{\text{W}}{\text{m K}} = \frac{\text{J}}{\text{m s K}} \right]$

Assume: 1) properties are constant (ρ, c_p, κ)

2) flow is incompressible $\nabla \cdot \underline{v} = 0$

substitute and simplify

$$\rho c_p \frac{\partial T}{\partial t} - \nabla \cdot [\underline{v} \rho c_p T - \kappa \nabla T] = \rho r \Rightarrow \text{HW9}$$

Advective heat flux: $q_A = \underline{v} \rho c_p T$

Conductive heat flux: $q_c = -\kappa \nabla T$

Heat production: ρr

$\Rightarrow$ Advection - Diffusion - Reaction Equation

Diffusion $\sim$ Conduction

Reaction $\sim$ Production

In limit of $\underline{v} \rightarrow 0$

$$\frac{\partial T}{\partial t} - D \nabla^2 T = \frac{r}{c_p} \quad \underline{\text{Heat Equation}}$$

where

$\nabla^2 = \nabla \cdot \nabla = \text{Laplacian}$ (Lecture 10)

$D = \frac{\kappa}{\rho c_p}$ thermal diffusivity $[\frac{m^2}{s}]$

$\kappa \rightarrow$ heat flux $D \rightarrow$ change in T

Example: Steady State Geotherm

Steady state: $\frac{\partial T}{\partial t} = 0 \Rightarrow -\kappa \nabla^2 T = \rho r$

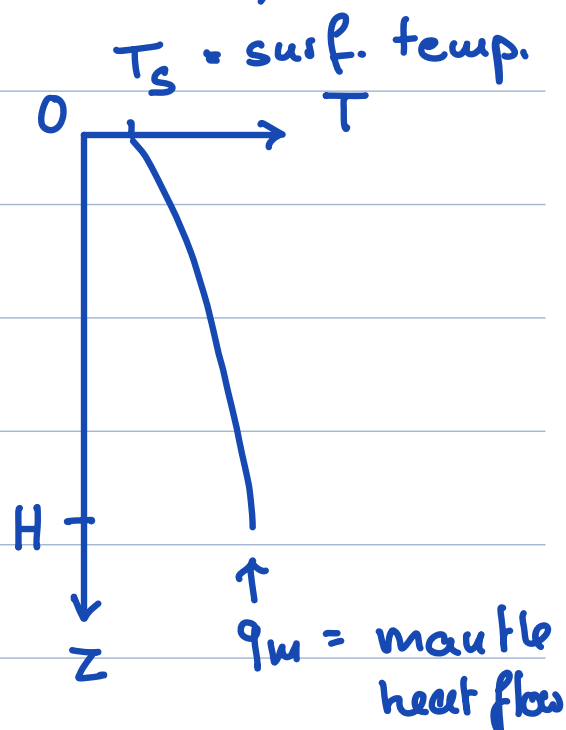
One dimension: $-\kappa \frac{d^2 T}{dz^2} = \rho r$

Boundary Conditions:

$$T(z=0) = T_s$$

$$q(z=H) = -\kappa \left. \frac{dT}{dz} \right|_{z=H} = -q_m$$

↑
upward!



Integrate once: $-\kappa \frac{dT}{dz^2} = \rho r \quad -\kappa \frac{dT}{dz} = \rho r z + c_1$

Use bottom BC: $-\kappa \left. \frac{dT}{dz} \right|_{z=H} = \rho r H + c_1 = -q_m$
 $\Rightarrow c_1 = -q_m - \rho r H$

$$\Rightarrow -\kappa \frac{dT}{dz} = \rho r z - \rho r H - q_m$$

Integrate again: $-\kappa T = \rho r \frac{1}{2} z^2 - (\rho r H + q_m) z + c_2$

Use top BC: $-\kappa T_s = c_2$

$$\Rightarrow -\kappa T = \frac{\rho r}{2} z^2 - (\rho r H + q_m) z - \kappa T_s$$

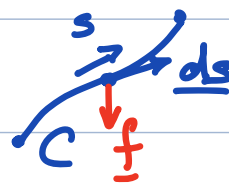
$$T(z) = T_s + \left(\frac{\rho r H}{\kappa} + \frac{q_m}{\kappa} \right) z - \frac{\rho r}{2\kappa} z^2$$

Work and Power

Work is energy transferred by application of force along distance

$$W = F s$$

$$W = \int_C \underline{f} \cdot d\underline{s} = \int_{t_1}^{t_2} \underline{f} \cdot \frac{d\underline{s}}{dt} dt = \int_{t_1}^{t_2} \underline{f} \cdot \underline{v} dt$$



Power is the rate of work

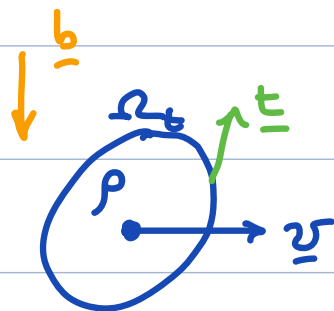
$$\mathcal{P} = \frac{dW}{dt} = \underline{f} \cdot \underline{v}$$

from

$$W = \int_{t_1}^{t_2} \frac{dW}{dt} dt = \int_{t_1}^{t_2} \underline{f} \cdot \underline{v} dt$$

Working is rate of work done $\sim$ power

Exchange of mechanical energy of a body



Kinetic Energy of a continuum body is

$$\mathcal{K}[\Omega_t] = \int_{\Omega} \frac{1}{2} \rho(\underline{x}, t) |\underline{v}(\underline{x}, t)|^2 dV_x$$

Power of external forces acting on Ω_t is

$$\mathcal{P}[\Omega_t] = \int_{\Omega_t} \rho(\underline{x}, t) \underline{b}(\underline{x}, t) \cdot \underline{v}(\underline{x}, t) dV_x + \int_{\partial\Omega_t} \underline{t}(\underline{x}, t) \cdot \underline{v}(\underline{x}, t) dA_x$$

Net working $\mathcal{W}[\Omega_t]$ of external forces on Ω_t is the mechanical power not converted into motion

$$\mathcal{W}[\Omega_t] = \mathcal{P}[\Omega_t] - \frac{d}{dt} \mathcal{K}[\Omega_t]$$

$\mathcal{W}[\Omega_t] > 0$: mechanical energy is stored

$\mathcal{W}[\Omega_t] < 0$: stored energy is released

Net working in Eulerian form

Power: $\mathcal{P} = \underline{f} \cdot \underline{v}$

Newton's 2nd law: $\underline{f} = m \underline{a} \rightarrow \underline{f} = \frac{d}{dt} (m \underline{v}) = m \underline{\dot{v}}$

Start by taking dot product of $\underline{v}$ and lin. mom. balance

lin. mom. balance: $\rho \underline{\dot{v}} = \nabla \cdot \underline{\underline{\underline{\sigma}}} + \rho \underline{b}$

$$\rho \underline{\dot{v}} \cdot \underline{v} = \rho \underline{v} \cdot \underline{\dot{v}} = (\nabla \cdot \underline{\underline{\underline{\sigma}}}) \cdot \underline{v} + \rho \underline{b} \cdot \underline{v}$$

Identify $\mathcal{K}$, $\mathcal{P}$ and $\mathcal{W} \Rightarrow$ integrate

$$\int_{\Omega_t} \rho \underline{v} \cdot \underline{\dot{v}} dV_x = \int_{\Omega_t} (\nabla_x \cdot \underline{\underline{\underline{\sigma}}}) \cdot \underline{v} + \rho \underline{b} \cdot \underline{v} dV_x$$

use identity $\nabla \cdot (\underline{A}^T \underline{b}) = (\nabla \cdot \underline{A}) \cdot \underline{b} + \underline{A} : \nabla \underline{b}$ (Lecture 10)

$$\int_{\Omega_t} \rho \underline{v} \cdot \dot{\underline{v}} dV_x = \int_{\Omega_t} -\underline{\underline{\underline{\sigma}}} : \nabla_x \underline{v} + \nabla \cdot (\underline{\underline{\underline{\sigma}}}^T \underline{v}) + \rho \underline{b} \cdot \underline{v} dV_x$$

Using $\underline{\underline{\underline{\sigma}}} : \underline{\underline{\underline{D}}} = \underline{\underline{\underline{\sigma}}} : \text{sym}(\underline{\underline{\underline{D}}})$ if $\underline{\underline{\underline{\sigma}}} = \underline{\underline{\underline{\sigma}}}^T$

$\Rightarrow$ introduce $\underline{\underline{\underline{d}}} = \text{sym}(\nabla_x \underline{v}) = \frac{1}{2}(\nabla_x \underline{v} + \nabla_x \underline{v}^T)$.

$$\int_{\Omega_t} \rho \underline{v} \cdot \dot{\underline{v}} dV_x = \int_{\Omega_t} -\underline{\underline{\underline{\sigma}}} : \underline{\underline{\underline{d}}} + \rho \underline{b} \cdot \underline{v} dV_x + \underbrace{\int_{\partial \Omega_t} \underline{\underline{\underline{\sigma}}} \underline{v} \cdot \underline{n} dA_x}_{\text{divergence thm.}}$$

Using $\underline{\underline{\underline{\sigma}}} \underline{v} \cdot \underline{n} = \underline{v} \cdot \underline{\underline{\underline{\sigma}}}^T \underline{n} = \underline{v} \cdot \underline{t}$

$$\int_{\Omega_t} \rho \underline{v} \cdot \dot{\underline{v}} dV_x = \int_{\Omega_t} -\underline{\underline{\underline{\sigma}}} : \underline{\underline{\underline{d}}} dV_x + \underbrace{\int_{\Omega_t} \rho \underline{b} \cdot \underline{v} dV_x + \int_{\partial \Omega_t} \underline{t} \cdot \underline{v} dA_x}_{\mathcal{P}[\Omega_t]}$$

Now we can identify the left hand side as

$$\frac{d}{dt} \mathcal{K}[\Omega_t] = \frac{d}{dt} \int_{\Omega_t} \frac{1}{2} \rho \underline{v} \cdot \underline{v} dV_x = \frac{1}{2} \int_{\Omega_t} \rho \frac{d}{dt} (\underline{v} \cdot \underline{v}) dV_x$$

$$\frac{d}{dt} (v_i v_i) = \dot{v}_i v_i + v_i \dot{v}_i = 2(v_i \dot{v}_i)$$

$$\frac{d}{dt} \mathcal{K}[\Omega_t] = \int_{\Omega_t} \rho \underline{v} \cdot \dot{\underline{v}} dV_x$$

so that we have the result

$$\frac{d}{dt} \mathcal{K}[\Omega_t] + \int_{\Omega_t} \underline{\underline{\sigma}} : \underline{\underline{d}} \, dV_x = \mathcal{P}[\Omega_t]$$

by comparison with $\mathcal{W}[\Omega_t] = \mathcal{P}[\Omega_t] - \frac{d}{dt} \mathcal{K}[\Omega_t]$

$$\Rightarrow \mathcal{W}[\Omega_t] = \int_{\Omega_t} \underline{\underline{\sigma}} : \underline{\underline{d}} \, dV_x$$

The quantity $\underline{\underline{\sigma}} : \underline{\underline{d}}$ is called the stress power associated with a motion. It corresponds to the rate of work done by internal forces (stresses) in a continuum body.

In some cases the power of an external force can be written as $\mathcal{P}[\Omega_t] = - \frac{d}{dt} G[\Omega_t]$

where $G[\Omega_t]$ is called potential energy for ext. forces.

$$\Rightarrow \frac{d}{dt} (\mathcal{U}[\Omega_t] + \mathcal{K}[\Omega_t] + G[\Omega_t]) = 0$$

Energy balance with net working

$$\frac{d}{dt} U[\Omega_t] = Q[\Omega_t] + W[\Omega_t]$$

$$\text{where } U[\Omega_t] = \int_{\Omega_t} \rho u \, dV_x$$

$$Q[\Omega_t] = \int_{\Omega_t} \rho r \, dV_x - \int_{\partial\Omega_t} \mathbf{q} \cdot \underline{n} \, dA_x$$

$$W[\Omega_t] = \int_{\Omega_t} \underline{\sigma} : \underline{d} \, dV_x$$

Hence we have

$$\frac{d}{dt} \int_{\Omega_t} \rho u \, dV_x = \int_{\Omega_t} \underline{\sigma} : \underline{d} \, dV_x - \int_{\partial\Omega_t} \mathbf{q} \cdot \underline{n} \, dA_x + \int_{\Omega_t} \rho r \, dV_x$$

using derivative relative to mass and divergence Thm

$$\int_{\Omega} (\rho \dot{u} - \underline{\sigma} : \underline{d} + \nabla_x \cdot \mathbf{q} + \rho r) \, dV_x$$

by the arbitrary ness of Ω_t we have

$$\underbrace{\rho \dot{u} = \underline{\sigma} : \underline{d} - \nabla_x \cdot \mathbf{q} + \rho r}_{\text{net working}} \quad \text{local Eulerian form}$$

$$\frac{\partial}{\partial t} (\rho u) + \nabla_x \cdot [\underline{v} \rho u + \mathbf{q}] = \underline{\sigma} : \underline{d} + \rho r \quad \text{conservative local Eulerian form}$$

Entropy and The Second Law

The Second Law expresses the fact that a body has a limit on the rate of heat uptake, but has no limit on the rate of heat release.

The second law postulates:

$$Q[\Omega_t] \leq \Xi[\Omega_t]$$

an upper bound $\Xi[\Omega_t]$ on the net heating. In the absence of net working $W[\Omega_t]=0$ we have

$$\frac{d}{dt} \mathcal{U}[\Omega_t] = Q[\Omega_t] \leq \Xi[\Omega_t]$$

$\Rightarrow$ the quantity $\Xi[\Omega_t]$ limits rate of energy storage.

The entropy of a body is defined (up to a constant)

$$\frac{d}{dt} S[\Omega_t] = \frac{\Xi[\Omega_t]}{\Theta[\Omega_t]}$$

where $\Theta[\Omega_t]$ is the uniform temp. of Ω_t .

Entropy is a quantity whose rate of change is equal to the upper heating bound per unit temperature.

At the atomistic level entropy is a measure of disorder, the number of atomic configurations compatible with macroscopic variables. In terms of net heating

$$\frac{d}{dt} S[\Omega_t] \geq \frac{Q[\Omega_t]}{\Theta[\Omega_t]}$$

Clausius-Planck
Inequality

In thermo books: $dS = \frac{dQ_{\text{rev}}}{T}$ and $dS > \frac{dQ}{T}$

The irreversibility of natural processes is shown by

$$\frac{d}{dt} S[\Omega_t] \geq 0 \quad \text{when} \quad Q[\Omega_t] = 0$$

For a non-homogeneous body we postulate an

entropy density field per unit mass, $s(x, t)$, so that

$$S[\Omega_t] = \int_{\Omega_t} \rho(\underline{x}, t) s(\underline{x}, t) dV_x$$

Hence the generalization of the 2nd law to inhomogeneous systems is given by the Clausius-Duhem ineq.

$$\frac{d}{dt} S[\Omega_t] \geq \int_{\Omega_t} \frac{\rho(\underline{x}, t) r(\underline{x}, t)}{\theta(\underline{x}, t)} dV_x - \int_{\partial\Omega_t} \frac{\underline{q}(\underline{x}, t) \cdot \underline{n}(\underline{x}, t)}{\theta(\underline{x}, t)} dA_x$$

This relation places restrictions on the constitutive relations of a body and leads to statements about energy dissipation and the flow of heat.

Note: The precise form of the 2nd law in continuum mechanics is not settled yet !

But all forms lead to same restrictions.

Local Eulerian Form of the Second Law

The integral form of the Clausius-Duhem form of the Second Law is

$$\frac{d}{dt} \int_{\Omega_t} \rho s \, dV_x \geq \int_{\Omega_t} \frac{\rho r}{\theta} \, dV_x - \int_{\partial\Omega_t} \frac{\mathbf{q} \cdot \mathbf{n}}{\theta} \, dA_x$$

After applying the Divergence Thm and invoking the arbitrariness of Ω_t we have

$$\rho \dot{s} \geq \rho r / \theta - \nabla_x \cdot (\mathbf{q} / \theta) \quad \text{Clausius-Duhem inequality in local Eulerian form}$$

After multiplying by θ and expanding the divergence

$$\theta \rho \dot{s} \geq \rho r - \nabla_x \cdot \mathbf{q} + \theta^{-1} \mathbf{q} \cdot \nabla_x \theta$$

Which can be written as

$$\mathcal{S} - \theta^{-1} \mathbf{q} \cdot \nabla_x \theta \geq 0$$

where $\mathcal{S} = \theta \rho \dot{s} - (\rho r - \nabla_x \cdot \mathbf{q})$ is the internal dissipation density per unit volume. Difference between local entropy increase and the local heating.

Note:

I) Any point where $\nabla_x \theta = 0$ the dissipation is non-negative, $S \geq 0$. $\Rightarrow$ bodies with homogeneous θ have non-neg. dissipation.

II) If $S = 0$, i.e. a reversible process, then $q \cdot \nabla_x \theta \leq 0$.

 Thus q is at an angle > 90 from $\nabla_x \theta$.

$\Rightarrow$ heat flows down the temperature gradient.

To study the consequences of Clausius-Duhem inequality for constitutive laws we introduce the field

$$\psi(x, t) = \phi(x, t) - \theta(x, t) s(x, t)$$

Helmholtz free energy density. This is the portion of the free energy available for performing work at const. θ .

$\Rightarrow$ Reformulate Clausius-Duhem in terms of ψ

Material derivative of free energy

$$\begin{aligned}\frac{d}{dt}(\theta s) &= \frac{\partial}{\partial t}(\theta s) + \nabla_x(\theta s) \cdot \underline{v} = \theta \frac{\partial s}{\partial t} + s \frac{\partial \theta}{\partial t} + \theta \nabla_x s \cdot \underline{v} \\ &\quad + s \nabla_x \theta \cdot \underline{v} \\ &= \theta \left(\frac{\partial s}{\partial t} + \nabla_x s \cdot \underline{v} \right) + s \left(\frac{\partial \theta}{\partial t} + \nabla_x \theta \cdot \underline{v} \right) \\ &= \theta \dot{s} + s \dot{\theta}\end{aligned}$$

from definition of $\dot{\psi}$

$$\dot{\psi} = \dot{u} - \theta \dot{s} - \dot{\theta} s \quad \Rightarrow \quad \dot{u} = \dot{\psi} + \theta \dot{s} + s \dot{\theta}$$

substituting into local form of 1st law

$$\rho \dot{u} = \underline{\underline{\sigma}} : \underline{\underline{d}} - \nabla_x \cdot \underline{q} + \rho r$$

$$\rho \dot{\psi} + \rho \theta \dot{s} + \rho s \dot{\theta} = \underline{\underline{\sigma}} : \underline{\underline{d}} - \nabla_x \cdot \underline{q} + \rho r$$

$$\rho \theta \dot{s} = \underline{\underline{\sigma}} : \underline{\underline{d}} - \nabla_x \cdot \underline{q} + \rho r - \rho \dot{\psi} - \rho s \dot{\theta}$$

substituting into 2nd law

$$\theta \rho \dot{s} \geq \rho r - \nabla_x \cdot \underline{q} + \theta^{-1} \underline{q} \cdot \nabla_x \theta$$

$$\underline{\underline{\sigma}} : \underline{\underline{d}} - \cancel{\nabla_x \cdot \underline{q}} + \cancel{\rho r} - \rho \dot{\psi} - \rho s \dot{\theta} \geq \cancel{\rho r} - \cancel{\nabla_x \cdot \underline{q}} + \theta^{-1} \underline{q} \cdot \nabla_x \theta$$

solve for $\rho \dot{\psi}$

$$\rho \dot{\psi} \leq \underline{\underline{\sigma}} : \underline{\underline{d}} - \rho s \dot{\theta} - \theta^{-1} \underline{q} \cdot \nabla_x \theta$$

This is called the reduced Clausius-Duhem inequality, because it is independent of local heat supply r and heat flux, q , if $\nabla_x \theta = 0$. $\Rightarrow$ homogeneous bodies of classical thermo.

Note:

In a homogeneous body, $\nabla\theta=0$, we have that

$$\rho \dot{\phi} \leq \underline{\sigma} : \underline{d}$$

for a reversible process this becomes an equality.

$\Rightarrow$ rate of change of free energy is equal to the stress power.